Tier 4 / Ecosystem

Compose services and negotiate extensions.

Tool design

Descriptions state scope, input constraints, effect and returned evidence.

Compare catalog size and task success before adding more tools.

An output schema validates structure; it does not prove factual accuracy.

Frameworks

Frameworks own orchestration; MCP owns the external capability boundary.

The TypeScript track runs Agents SDK, LangChain and LangGraph against one read-only workflow.

Keep model/runtime versions and approval policy explicit.

Gateway

Use fixed upstream aliases and collision-free names. Partition catalogs/caches by principal and scopes.

Recheck permissions at execution; limit rates and record redacted audit events.

A failed upstream should not silently expose hidden tools or block unrelated safe reads.

Tasks

Advertise io.modelcontextprotocol/tasks before accepting a task handle.

Poll tasks/get; answer input with tasks/update; cancel with tasks/cancel.

Persist ownership and expiry. Bound polling. A current Task handle is not the legacy SDK task shape.

Apps and Skills

Apps link a tool to a UI resource and a controlled bridge. Enforce browser sandbox/CSP and a narrow tool allowlist.

Skills are discovered and byte-verified before a separate activation decision. Origin plus URI identifies a skill.

Checkpoint

uv run python scripts/tier4.py

uv run python scripts/gateway_lab.py

Next: Gateway capstone and shipping modules 18-21.